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Paleomagnetic data and U-Pb isotopic age determinations from Coats Land, Antarctica: Implications for late Proterozoic plate reconstructions

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Abstract. Paleomagnetic results and isotopic age determinations for granophyre and rhyolite from small, isolated nunataks in southern Coats Land, Antarctica, are used to evaluate late Proterozoic plate reconstructions. U-Pb zircon dates for the two rock types indicate coeval crystallization at 1112 ± 4 Ma. A concordant 1106 ± 3 Ma titanite date from the granophyre overlaps the crystallization age, implying rapid cooling, and is consistent with field and petrographic evidence of no subsequent penetrative deformation, metamorphism, or hydrothermal disturbance. The mean direction of magnetization of the rhyolite at Littlewood Nunataks is statistically indistinguishable from the mean directions of five sites in the granophyre and crosscutting rhyolite dikes at Bertrab Nunataks. The group mean virtual geomagnetic pole of 22.9°N, 80.3°E (N=6, A_{95}=6.8°) compares favorably with the only other extant Precambrian paleomagnetic poles for the East Antarctic craton, two poles from western Dronning Maud Land. The East Antarctic and Laurentian poles of 1.1 Ga do not coincide after restoration of the continents to a position suggested by the SWEAT hypothesis juxtaposing the Pacific margins of East Antarctica-Australia and Laurentia, indicating either that the hypothesis is incorrect or that Coats Land and parts of western Dronning Maud Land (herein the CMG province) were not part of the East Antarctic craton at 1.1 Ga. In support of the latter, there is reasonable agreement of the 1.1 Ga CMG poles and approximately coeval poles from the Kalahari craton of West Gondwana when the CMG is restored to a position adjacent the Kalahari craton. Such a reconstruction places the CMG in West Gondwana rather than East Gondwana, as originally implied in Rodinia reconstructions, and is consistent with previously recognized links between the geology of the Kalahari craton and western Dronning Maud Land. It further implies that the CMG did not become part of East Antarctica until latest Precambrian to Cambrian time. A new reconstruction places a partially assembled West Gondwana off the present southeastern margin of Laurentia at 1.1 Ga such that poles of the CMG and Kalahari fall on the Laurentian polar wander path for this time period.

Introduction

Paleomagnetic data, Proterozoic and early to mid-Paleozoic ophiolitic complexes, and late Precambrian rifted margins all indicate that the Pangea supercontinent reconstructed using seafloor spreading data was not the original configuration of continents on Earth. Thermal subsidence analyses of late Precambrian rifted continental margins [*Bond et al.*, 1984] and comparison of apparent polar wander paths [*Piper*, 1987] suggest that at least one earlier supercontinent existed during the Neoproterozoic Era between the end of the Grenville orogeny and the deposition of lower Paleozoic miogeoclinal strata on all continents, i.e., between circa 1000 and 550 Ma. Until recently, however, the configuration of this supercontinent has remained obscure. *Moores* [1991], following earlier suggestions by Canadian workers [*Eisbacher*, 1985; *Bell and Jefferson*, 1987], juxtaposed the Laurentian and East Antarctic-Australian cratons prior to the opening of the Pacific Ocean basin near the end of the Precambrian. He referred to this as the "SWEAT" hypothesis on the basis of intercratonic connections he believed could be identified between the southwest United States. and East Antarctica.

Most critically, *Moores* [1991] noted that igneous and metamorphic rocks in the Shackleton Range along the Pacific margin of the Precambrian craton of East Antarctica (Figure 1) have an age range of 1.6-1.8 Ga, comparable to those of the Yavapai and Mazatzal provinces of Arizona and New Mexico [*Karlstrom and Bowring*, 1988; *Hoffman*, 1989]. He also pointed to the existence in Dronning Maud Land, that part of

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the East Antarctic craton south of Africa (in present-day coordinates; Figure 1), of high-grade metamorphic rocks with an age range comparable to that of the Grenville province of eastern and southern North America, circa 1.2-1.0 Ga. He further suggested that the "Grenvillian" rocks of Dronning Maud Land can be traced along strike into southeastern India and the Albany-Fraser belt of southwestern Australia, thereby implying that East Gondwana was assembled by the end of the Grenville orogeny.

Dalziel [1991] supported *Moores'* [1991] hypothesis with a computer-generated reconstruction and pointed out the existence of extremely small, isolated outcrops of silicic volcanic and hypabyssal rocks of Grenvillian age in southern Coats Land at the head of the Weddell Sea, 300 km from the Shackleton Range (Figure 1). *Dalziel* [1991, 1992a] used the location of these outcrops, herein referred to as the Coats Land Nunataks, to constrain the position of an Antarctic "Grenville Front" (Figure 1) to the ice-covered region between these nunataks and the Shackleton Range. The southwestern terminus of this front was suggested as a piercing point to be matched with the southern end of its Laurentian counterpart in the southwest United States, providing a major constraint for the SWEAT reconstruction. *Dalziel* [1991, 1992a, b, 1994] and *Hoffman* [1991] also suggested that the proto-Appalachian margin of Laurentia was juxtaposed with the proto-Andean margin of Amazonia during the Neoproterozoic, thus leading to new reconstructions of the supercontinent that was assembled during Grenvillian orogenesis (for a review, see *Dalziel* [1995]). This supercontinent, which would have existed prior to the opening of both the Pacific and Iapetus ocean basins, has been named Rodinia by *McMenamin and McMenamin* [1990]. A second supercontinental entity may have existed ephemerally following the opening of the Pacific Ocean and amalgamation of Gondwanaland and prior to the opening of Iapetus [*Dalziel*, 1992a, b, 1994]. This has been named Pannotia [*Powell*, 1995; *Dalziel*, 1997]. The paleogeography during the late Mesoproterozoic and Neoproterozoic eras is particularly important to understanding of Earth history. It marks a time of increased diversification and turnover rates in protistan microfossils and the emergence of Metazoan life, and immediately preceded the Cambrian "explosion" of marine invertebrates [*Knoll*, 1994].

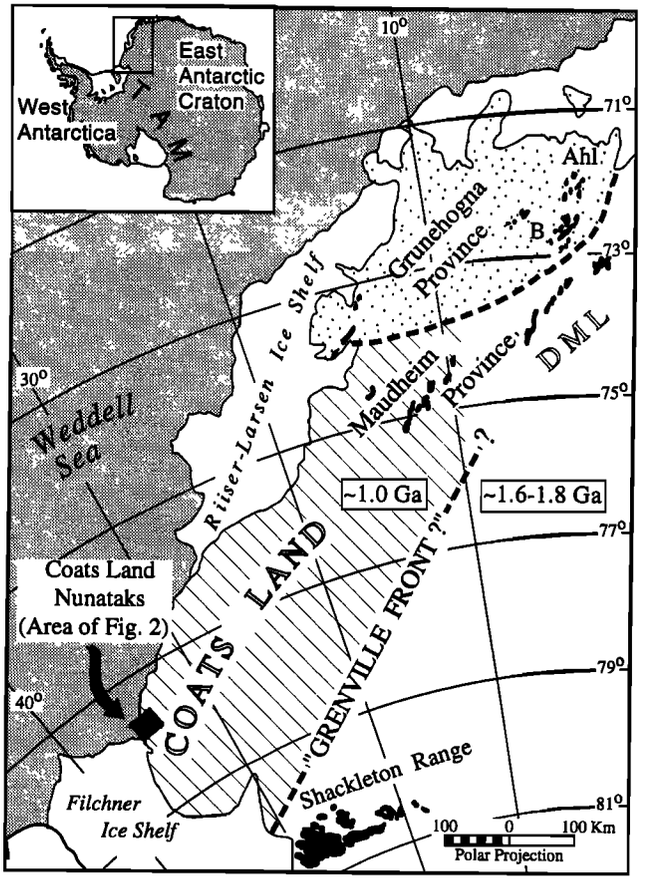


Figure 1. Map of outcrops of Precambrian and lower Paleozoic rocks along the Weddell Sea margin of the East Antarctic Precambrian craton (see inset), showing the suggested continuation of the Grenville Front of Laurentia [*Dalziel*, 1992b], the Maudheim and Grunehogna provinces [from *Moyes et al.*, 1993], the Shackleton Range, and the location of the Coats Land nunataks. Generalized areas of rock exposure are shown in black. Abbreviations are Ahl, Ahlmannryggen; B, Borgmassivet; DML, Dronning Maud Land; TAM, Transantarctic Mountains.

Meert et al. [1994] used paleomagnetic poles from several continents to demonstrate that the Rodinian reconstruction of *Dalziel* [1992a] could not have existed at circa 1250 Ma, prior to the Grenville Orogeny. However, *Powell et al.* [1993] showed that Neoproterozoic poles are compatible with juxtaposition of the Pacific margins of Laurentia and East Gondwana during the period circa 900-725 Ma. The data from East Gondwana used in their analysis, however, come from Australia and India, and the timing of East Gondwana amalgamation is not well established [*Grunow et al.*, 1996]. For example, there may be an early Paleozoic suture in the Shackleton Range [*Marsh*, 1983; *Buggisch et al.*, 1993; *Dalla Salda et al.*, 1992b; *Tessensohn*, 1996] and along the Indian Ocean margin south of India [*Shiraishi et al.*, 1994]. A more thorough test of the SWEAT hypothesis therefore requires Neoproterozoic and/or late Mesoproterozoic paleomagnetic data from the East Antarctic craton. Owing to the extent of the East Antarctic ice sheet, exposed rocks suitable for such analyses are restricted to the periphery of the craton, a position that may render them suspect and unrepresentative of the craton proper due to potential displacements or translations, especially for exposures along the Transantarctic Mountains [e.g., *Borg and DePaolo*, 1994]. Yet these rocks represent the best, and in most cases the only, opportunity for testing the hypothesis. Even if such rocks prove to be allochthonous and unsuitable for testing SWEAT per se, they may reveal, as discussed below, important insights into the paleogeography of this important time interval that are unobtainable elsewhere.

Here we report the results of paleomagnetic and geochronologic studies of the circa 1.1 Ga igneous rocks of the Coats Land Nunataks, the rocks suggested by *Dalziel* [1991] to be the most proximal equivalents in East Antarctica of the Grenville province of the southern United States. These data indicate, however, that prior to circa 550 Ma, Coats Land was part of West Gondwana rather than East Gondwana. Thus the "Antarctic Grenville front" cannot be used to define the relative positions of the Laurentian and East Antarctic cratons. The data do, however, contribute to a better understanding of Neoproterozoic paleogeography by suggesting proximity of the Laurentian and Kalahari cratons at circa 1100 Ma, and demonstrate that the East African suture between East and West Gondwanaland may have extended to the Pacific margin of

Antarctica as recently suggested by *Grunow et al.* [1996] and *Tessensohn* [1996].

Regional Setting

Grenville age rocks (1.2-1.0 Ga) in western Dronning Maud Land and Coats Land are found in the amphibolite- to granulite-facies Maudheim Province [*Groenewald et al.*, 1991] and some 600 km to the southwest in the small, isolated nunataks of southern Coats Land (Figure 1). Geologic similarities and early Mesozoic continental reconstructions controlled by seafloor spreading data [*Martin and Hartnady*, 1986; *Lawver and Scotese*, 1987] suggest that the Maudheim Province and the bordering Archean basement and Mesoproterozoic cover to the north, the Grunehogna Province (Figure 1) [*Groenewald et al.*, 1991], were contiguous with the Natal/Namaqua/Mozambique Belts and the Kalahari craton, respectively, in southern Africa [*Moyes et al.*, 1993; *Jacobs et al.*, 1993; *Groenewald*, 1993] until Mesozoic break up of Gondwanaland. With the exception of the stable interior of the Kalahari craton, all of these areas underwent varying degrees of tectonic and metamorphic reworking in latest Proterozoic to Early Cambrian time during the Pan African/Brazilide orogenies of interior Gondwanaland and the Ross orogeny of the Transantarctic (paleo-Pacific) margin. In contrast, the nunataks of southern Coats Land apparently escaped such overprinting [*Aughenbaugh et al.*, 1965; *Marsh and Thomson*, 1984; *Storey et al.*, 1994], making them among the best candidates in this part of the East Antarctic craton for paleomagnetic study.

Coats Land Nunataks

The Coats Land Nunataks comprise three, very small, isolated groups of exposures of igneous rocks, the Moltke, Bertrab, and Littlewood Nunataks (Figure 2), of which only the latter two are readily accessible; we found the rocks at the Moltke locality to be inaccessible beneath a vertical ice fall.

Bertrab Nunataks

The Bertrab Nunataks are composed of red- to gray-weathered, fine- to medium-grained, oligoclase-phyric granophyre. The granophyre is cut by 1-7 m wide, flow-banded rhyolite dikes and 1-2 m wide, fine-grained mafic dikes (Figure 2c). Detailed petrographic descriptions of the granophyre and dikes at the largest nunatak (at the Argentine Belgrano II base) are provided by *Toubes Spinelli* [1983] and *Marsh and Thomson* [1984]. *Storey et al.* [1994] reported a Rb-Sr whole rock isochron age of 1076 ± 7 Ma (mean square of weighted deviates (MSDW) = 1.6) for eight samples of granophyre, rhyolite and diabase from this nunatak and suggest, on the basis of these and other limited geochemical data, that the dikes and granophyre are cogenetic. The rocks of the previously undescribed other two nunataks of the Bertrab group are nearly identical to those of the largest exposure, except that the granophyre is locally slightly coarser- and/or finer-grained. The middle nunatak of the group is cut by an ~2 m wide diabase dike that is in turn cut by a 1 m wide, flow-banded rhyolite dike identical to those at the larger nunatak but trending 045°. Granophyric textures like those found throughout the Bertrab Nunataks are indicative of rapid crystallization and/or devitrification and are commonly associated with hypabyssal or extrusive modes of emplacement [cf. *Barker*, 1970].

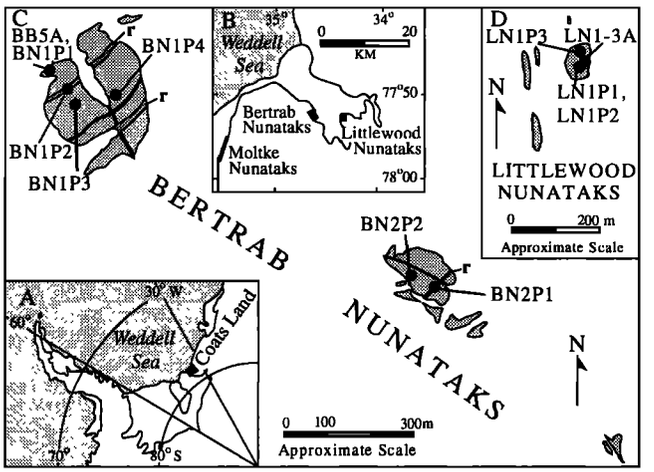


Figure 2. Maps of the Bertrab and Littlewood Nunataks showing the location, geology, and sampling sites. The location of Figure 2b is shown by the black box in Figure 2a. Figure 2b shows the location of the Bertrab of Littlewood Nunataks. Medium shading in Figures 2c and 2d denotes granophyre at the Bertrab Nunataks and rhyolite at the Littlewood Nunataks. Within the Bertrab Nunataks, solid, black lines are mafic dikes and NE trending dikes labeled "r" are rhyolite.

Paleomagnetic samples were collected at six sites in the Bertrab Nunataks; four from granophyre (BN1P1, BN1P3, BN1P4, and BN2P2) and two from rhyolite dikes (BN1P2, BN2P1) (Figure 2c). Granophyre samples are composed of a fine-grained, granophyric matrix of quartz and alkali feldspar, which hosts fine- to medium-grained phenocrysts of highly zoned plagioclase, and irregularly dispersed concentrations of biotite, magnetite, titanite, chlorite, and, rarely, amphibole, apatite, and zircon. The granophyre is extremely fresh and unweathered in outcrop. Thin section examination reveals moderate to thorough sericitization of plagioclase and microtextures and paragenetic relationships among the mafic minerals and oxides that are indicative of hydrothermal and/or deuteric alteration. Relict amphibole is partially or completely replaced by biotite, chlorite, and magnetite, and biotite is partially replaced by chlorite. Magnetite is rimmed by anhedral, extremely fine-grained, yellow titanite, which together are spatially associated with clusters of amphibole and biotite. The yellow titanite is morphologically distinct from coarser-grained, isolated, brown, subhedral titanite. The brown titanite is not spatially associated with mafic mineral clusters, nor with partially reacted mineral phases, and is thus interpreted as primary.

The two sampled rhyolite dikes are vertical, flow banded at their margins, and range from 1 to 2.5 m in width. Both are composed of a microcrystalline groundmass of quartz and feldspar that contains sparse phenocrysts of quartz, microcline microperthite, biotite, and, rarely, plagioclase. The groundmass locally exhibits spherulitic and granophyric textures. Very fine-grained to fine-grained magnetite is dispersed throughout the matrix and feldspar phenocrysts. The rhyolite dikes do not show petrographic evidence for alteration like that seen in the granophyres.

Littlewood Nunataks

This second group of outcrops comprises five small exposures of red-weathered, densely silicified rhyolite that are located about 10 km east of the Bertrab Nunataks (Figure 2d). *Aughenbaugh et al.* [1965] provided lithologic descriptions, a chemical analysis, and a whole rock K-Ar age of 851 ± 3 0 Ma (recalculated following *Dalrymple* [1979]) for rhyolite at the largest outcrop of the group. The significance of this age is unclear. Volcanic rocks, particularly those that are devitrified, generally do not behave as closed systems with respect to argon and may yield ages that can be substantially younger than the age of crystallization or cooling determined by other methods [*McDougall and Harrison*, 1988]. The only other published age determination for Littlewood rhyolite is a 1001 ± 16 Ma whole rock Rb-Sr isochron for a mixed suite of Littlewood and Bertrab samples [*Eastin and Faure*, 1971]. When recalculated with modern decay constants and more conservative error estimates, the same data yield an age of 976 ± 35 Ma [*Storey et al.*, 1994]. Despite differences in isotopic ages, Littlewood rhyolite and Bertrab granophyres and dikes have been previously been regarded as coeval and cogenetic on the basis of limited isotopic, major and trace element data [*Eastin and Faure*, 1971; *Storey et al.*, 1994].

Paleomagnetic samples of Littlewood rhyolite were collected at three sites (LN1P1, LN1P2, and LN1P3) in the largest Littlewood outcrop (Figure 2d). Rhyolite at this locality is composed of microcrystalline quartz and feldspar that contains sparse, angular fragments of sanidine and quartz and is pervaded by chalcedony-filled or, more rarely, drusy quartz-lined veinlets. Black to gray, angular, volcanic rock fragments up to 2 cm in longest dimension are present in some samples. Thin section and scanning electron microscope examination shows that the opaque mineralogy of the rhyolite is dominated by hematite ranging from ~0.5 mm to 2 mm in largest dimension. Hematite is ubiquitous throughout the microcrystalline quartz and feldspar matrix, occurring both as isolated grains and crystal clusters, some of the latter having partially or completely replaced biotite.

The remaining Littlewood Nunataks, for which there are no previously published descriptions, are also composed of red-weathered, silicified rhyolite. Unlike the largest nunatak, however, rhyolite in these exposures locally preserves features in both outcrop and thin section that indicate a pyroclastic origin. Such features include the (1) common occurrence of angular volcanic lithic clasts, (2) angular, ragged, fragmentary appearance of feldspar phenocrysts, (3) rare occurrence of devitrified glass shards, and (4) presence of elongate, aligned, irregular cavities interpreted to be voids left by weathering of partially welded pumice. The textural characteristics of the rhyolite in all outcrops are similar to those of partially welded ash flow tuffs that have been altered and silicified by extensive vapor phase alteration, indicating these rocks underwent rapid deposition and cooling.

An indistinct, discontinuous layering defined by weathered surfaces and alignment of elongate cavities at the sampling sites and in other nunataks of the group has a strike that ranges between 020° and 028° and dips 10°-11° SE. On the basis of the textures described above, we interpret the layering to be primary and of ash flow origin. Ash flows mantle topography such that layering need not be horizontal. Lacking field and petrographic evidence to indicate subsequent deformation, we interpret the 10-11° dip as a primary feature.

U-Pb Geochronology

Precise dating of the crystallization and alteration history of the Littlewood and Bertrab rocks is crucial for accurate interpretation of the paleomagnetic data. Both units and the dikes that cut the granophyre have previously been regarded as cogenetic and coeval, but extant isotopic ages are not precise enough to adequately test this premise, nor do they address the age of alteration. Zircons in Littlewood rhyolite and Bertrab granophyre permit U-Pb dating of the crystallization of these rocks, and titanite in the granophyre can be used to date the alteration. Results of the U-Pb analyses of these minerals are discussed below and presented in Table 1. Analytical methods are given in the appendix.

Littlewood Rhyolite

Three fractions of high-quality zircons were analyzed from Littlewood rhyolite (sample LN13A, Figure 2d). Two fractions of approximately 50 euhedral, colorless, inclusion-free, needle-shaped grains plot within error on concordia with an average $^{207}Pb/^{206}Pb$ age of 1112 ± 4 Ma (Figure 3a and Table 1). Fraction Z3, composed of an equivalent number of stubbier, prismatic crystals with tiny (1-2 mm) clear inclusions, is slightly discordant (1.6%), giving a $^{207}Pb/^{206}Pb$ age of 1118 Ma (Figure 3a). The zircon morphology of all fractions is indicative of an igneous origin and the average 1112 ± 4 Ma $^{207}Pb/^{206}Pb$ age of the two concordant fraction thus dates the time of igneous zircon crystallization. We infer this age to be contemporaneous with the age of tuff emplacement. The discordance of Z3 is not well understood but may reflect the combined effects of a very minor component of older, inherited zircon and recent lead loss.

Bertrab Granophyre

Zircons from Bertrab granophyre (sample BB5a, Figure 2c) are of poorer quality than those from Littlewood rhyolite. Three fractions of near-equant, faceted zircons containing small inclusions and cracks were analyzed and define a discordia line with upper and lower intercepts of 1113 +18/-6 Ma and 212 ± 350 Ma respectively (92% probability of fit, where 10% represents a MSWD of 2) (Table 1 and Figure 3b). The upper intercept, which overlaps the age determined for Littlewood rhyolite, is interpreted to represent the time of crystallization of the granophyre. The 3-4% discordance of the best zircons is consistent with the lower average quality of these grains compared to those from the rhyolite and precludes determination of more precise intercept errors.

Two fractions of titanite, one of primary brown titanite (T1) and another of yellow secondary titanite (T2), were also analyzed (Table 1). Both fractions yield analytically indistinguishable results that overlap concordia and give an average $^{207}Pb/^{206}Pb$ age of 1106 ± 3 Ma (Figure 3b). The coincident age of both yellow and brown titanite suggests either that the texturally earlier brown titanite was completely reset by alteration that produced yellow titanite and magnetite from Fe-Ti oxides and silicates, or that this alteration occurred essentially contemporaneous with crystallization and is thus deuteric in origin. The latter interpretation is favored by the overlap of the titanite and zircon ages. In either event, the coincidence and concordance of both titanite analyses demonstrate that alteration of the granophyre is no younger than circa 1106 Ma.

Table 1. U-Pb Isotopic Data and Ages

Fraction	Concentration Measured				Corrected Atomic Ratios*								Ages, Ma					
	Weight mg	U	Pb rad ppm	total common Pb, pg	$\frac{^{206}Pb}{^{204}Pb}$	$\frac{^{208}Pb}{^{206}Pb}$	$\frac{^{206}Pb}{^{238}U}$		$\frac{^{207}Pb}{^{235}U}$		$\frac{^{207}Pb}{^{206}Pb}$		$\frac{^{206}Pb}{^{238}U}$		$\frac{^{207}Pb}{^{235}U}$		$\frac{^{207}Pb}{^{206}Pb}$	
LN13A: Littlewood Rhyolite																		
Z1 b euh	0.062	154	29.5	6	18712	0.1147	0.18727	84	1.9800	90	0.07668	10	1107	4.9	1109	5.0	1113	1.5
Z2 clr euh	0.050	133	25.8	10	7894	0.1263	0.18714	74	1.9770	78	0.07662	16	1106	4.3	1108	4.3	1111	2.3
Z3 65 sm clr rnd	0.026	137	26.2	8	5158	0.1134	0.18647	74	1.9757	82	0.07684	14	1102	4.3	1107	4.6	1117	2.0
BB5A: Bertrab Granophyre																		
Z1 b euh w cracks	0.045	178	33.4	5	16981	0.1103	0.18396	52	1.9420	62	0.07657	8	1089	3.1	1096	3.5	1110	1.2
Z2 sm eud-sub clr clrs	0.042	190	35.7	15	5995	0.1112	0.18351	54	1.9367	62	0.07654	8	1086	3.2	1094	3.5	1109	1.2
Z3 2nd b from Z2	0.024	199	37.0	5	11624	0.1154	0.18187	54	1.9182	62	0.07650	8	1077	3.2	1087	3.5	1108	1.2
T2 brn clr euh frags	0.186	79	17.7	197	881	0.3253	0.18615	124	1.9617	138	0.07643	14	1100	7.3	1102	7.7	1106	2.0
T3 yel clr frag	0.152	63	14.6	140	812	0.3616	0.18602	86	1.9589	104	0.07637	18	1100	5.1	1101	5.8	1105	2.6

Abbreviations are b, best; brn, brown; clr, clear; clrs, colorless; euh, euhedral; frags, fragments; rnd, rounded; sm, small (<50 μm); sub, subhedral; w, with; yell, yellow. Mineral codes are T, titanite; Z, zircon. Two-sigma errors are reported for ages.

*Ratios corrected for fractionation, 1-5 pg laboratory Pb blank, initial common Pb calculated using Pb isotopic compositions of *Stacey and Kramers* [1975] and 2 pg U laboratory blank. Two-sigma uncertainties are reported for isotopic ratios, calculated with a modified unpublished error propagation program written by L. Heaman. Errors reported after the ratios refer to the final digits. All fractions were strongly abraded [*Krogh*, 1982].

The overlap within errors of the zircon and titanite dates is also consistent with the rapid crystallization and cooling inferred from the rock textures. If the titanite age is regarded as a cooling age then the estimated 500-600°C closure temperature of titanite [*Tucker et al.*, 1987; *Heaman and Parrish*, 1991] requires that the granophyre initially cooled to or below the magnetite Curie temperature (580°C) by ca. 1106 Ma. The concordance of the titanite analyses and the absence of significant lead loss suggest that these rocks have not experienced significant reheating since emplacement, an inference consistent with the igneous textures and mineralogies and the absence of deformational fabrics. The 3-4% discordance of zircons from the granophyre can be attributed to recent lead loss that was enhanced by slightly higher U contents (Table 1) and the ubiquitous fractures within these grains. The highly imprecise lower intercept is within the range of Mesozoic events related to Gondwana breakup, which elsewhere include uplift, denudation, and emplacement of flood basalts (for review see *Dalziel* [1992a]). Any or all of these events could have promoted lead loss from these fractured zircons.

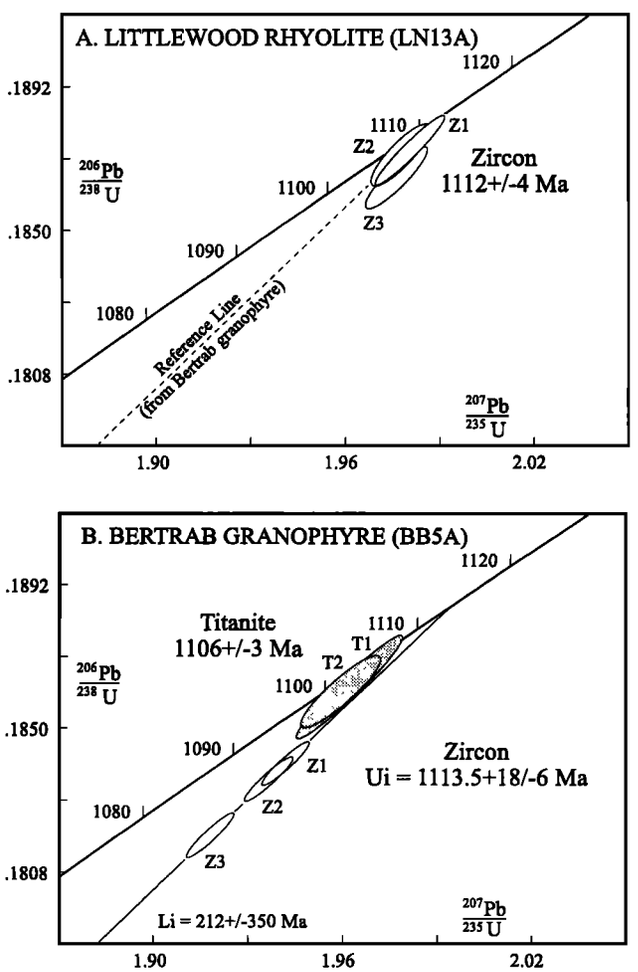


Figure 3. Concordia diagrams for (a) zircon from Littlewood rhyolite and (b) zircon and titanite (shaded) from Bertrab granophyre. Zircon and titanite fractions are labeled with reference to Table 1. Dashed reference line in Figure 3a is from the best fit discordia line for Bertrab granophyre zircons shown in Figure 3b. Abbreviations are Li, lower intercept; Ui, upper intercept.

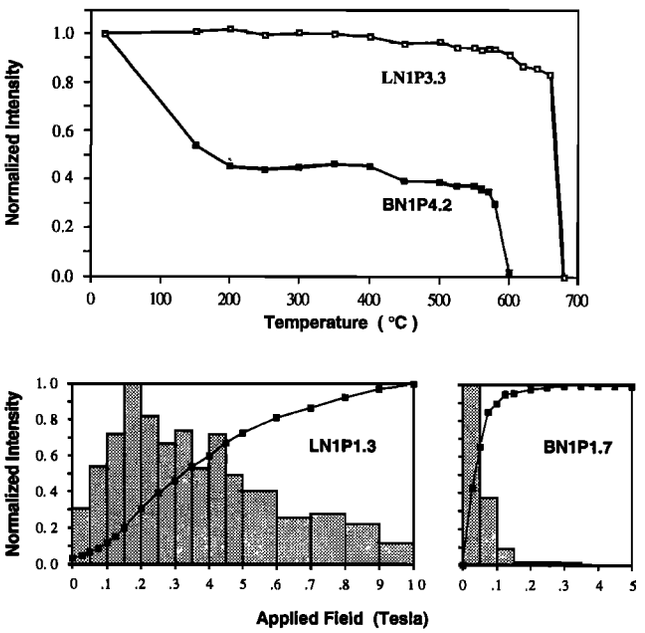


Figure 4. Thermal demagnetization and IRM acquisition with coercivity spectra for samples of Bertrab granophyre and Littlewood rhyolite. The analyses demonstrate that the magnetization in Bertrab samples is carried by magnetite and that Littlewood rhyolite contains hematite as the only magnetic mineral.

U-Pb dates from the rhyolite and granophyre demonstrate they are coeval. We consider the more precise U-Pb date of 1112 ± 4 Ma as representative of the crystallization age of this suite. A concordant date of 1106 ± 3 Ma for titanite from the granophyre overlaps the crystallization age, implies rapid cooling and is consistent with field and petrographic evidence for no subsequent thermal or tectonic disturbance.

Paleomagnetic Analysis

Samples for paleomagnetic analysis were collected from rhyolite at the largest Littlewood nunatak and from granophyre and rhyolite dikes at two of the Bertrab nunataks (Figure 2). Sampling procedures and analytical methods are described in the appendix.

Polished thin section and scanning electron microscope examinations identified magnetite in Bertrab samples and hematite in Littlewood samples. These observations are confirmed by isothermal remanent magnetization (IRM) acquisition experiments and coercivity spectrum analyses [*Dunlop*, 1972]. Bertrab samples acquire most of their remanence in applied fields of less than 0.1 mT and reach saturation by 0.3 mT (Figure 4). This behavior is indicative of a mixture of multi- and single-domain magnetite. The thermal demagnetization of the natural remanent magnetizations (NRMs) also shows two types of magnetite: one has unblocking temperatures ranging from 200° to 400°C, depending on the specific sample, and is interpreted as multidomain magnetite; the other has unblocking temperatures very near the magnetite Curie temperature (580°C), indicative of single-domain magnetite with very little or no titanium.

In contrast, Littlewood samples have hematite as the sole carrier of the magnetic remanence. The shape of the IRM

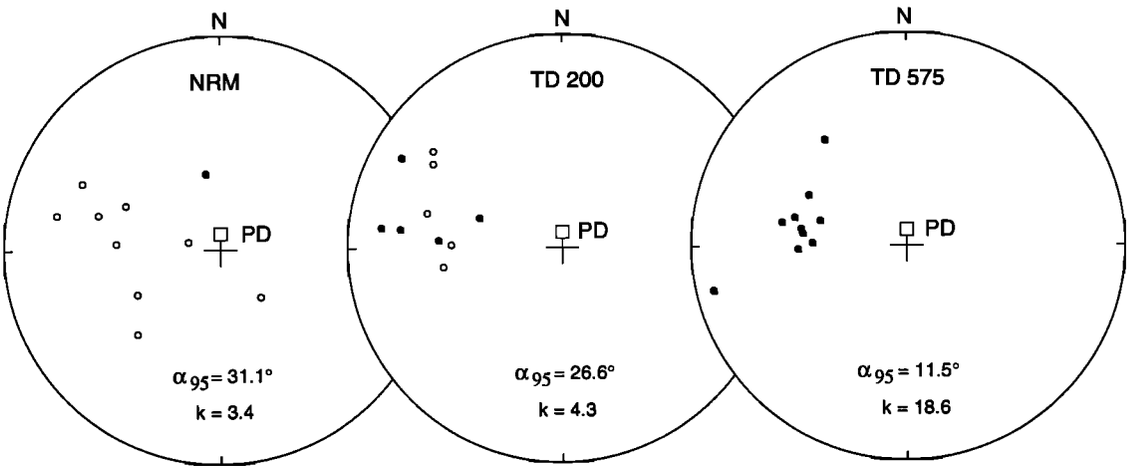


Figure 5. Equal-area stereographic projections of the directions of magnetization of the NRM and directions after thermal demagnetization at 200° and 575°C for samples from site BN1P1 in the Bertrab Nunataks. Open (solid) symbols are in the upper (lower) hemisphere. PD is the present dipole field direction, which has a negative inclination in the southern hemisphere. Same convention applies to all similar figures.

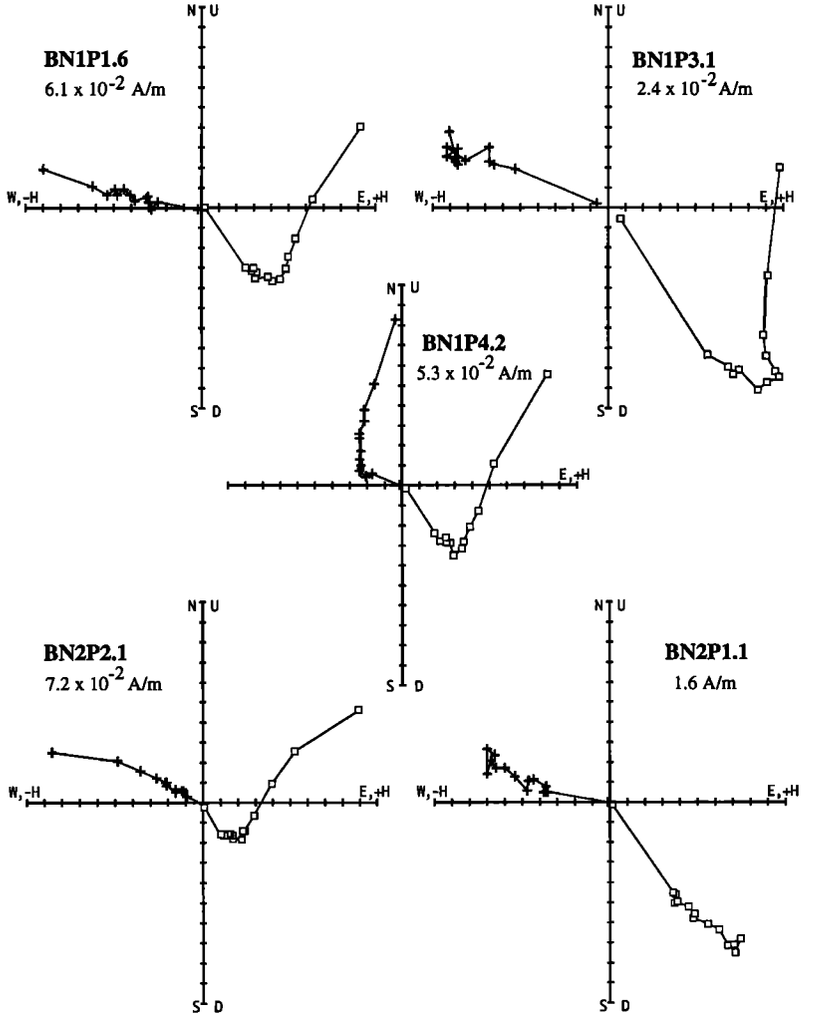


Figure 6. Vector component diagrams for Bertrab samples. Open squares are the projection onto the up-down-horizontal plane and crosses represent the projection onto the N-S-E-W plane. Scale is given below sample number. Almost all samples contain two component of magnetization, a low temperature component closely aligned with the present geomagnetic field and a well-defined, high-temperature component. Samples were thermally demagnetized at 150°, 200°, 250°, 300°, 350°, 400°, 450°, 500°, 525°, 550°, 560°, 570°, 580°, and 600°C.

acquisition curve (Figure 4) indicates the presence of coarse-grained hematite (specularite) as well as very fine-grained hematite. In thermal demagnetization of their NRMs, these samples retained over 80% of their remanence up to 660°C but lost most of their remanence after heating to 680°C, confirming hematite as the sole carrier of the remanent magnetization.

The response of Bertrab samples to thermal demagnetization is well illustrated by the behavior of the samples from BN1P1 (Figure 5). NRM directions are strongly biased towards the dipole field direction (declination is 0°, inclination is -84°) and move steadily toward a direction with a positive inclination. At the same time, the scatter of the directions decreases, as manifest by a decrease in α_{95} and an increase in Fisher's precision parameter k. Upon exceeding the magnetite Curie temperature, the directions disperse.

Most Bertrab samples contain two well-defined components of magnetization (Figure 6). One component is stable up to ~400°C and scatters around the present magnetic field direction (Figure 7). This magnetization is interpreted as a viscous magnetization residing in multidomain magnetite. In the samples from the rhyolite dike at site BN1P2, this was the only component, and these results are not considered further. The second component was isolated over a demagnetization range of 400° to 580°C. The linear decay toward the origin implies that no other high-temperature magnetization is present in these samples. Statistical parameters and results for the Bertrab samples are summarized in Table 2. The direction of magnetization of four samples differed from their site mean by more than two angular standard deviations; these samples have been deleted from the statistical analyses.

The hematite-bearing Littlewood samples responded very uniformly to thermal demagnetization. The directions retained a tight grouping throughout the entire demagnetization process up to 660°C. Because of the very narrow unblocking temperature range, the vector component diagrams (Figure 8) do not show linear trends but rather a cluster of points and the principal component method can not be applied. The characteristic direction of magnetization for each sample has been calculated using the Fisher mean of all demagnetization steps up to 660°C; statistical parameters are listed in Table 2.

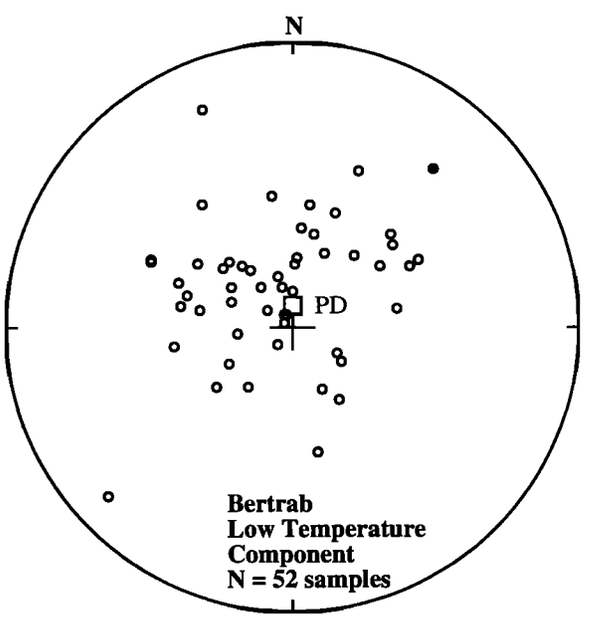


Figure 7. Equal-area projection showing that the directions of magnetization of the low-temperature component of Bertrab samples is strongly biased toward the present field direction (PD).

Interpretation of Paleomagnetic Data

The means from the three Littlewood sites are almost identical (Figure 9), which suggests that they are from a single rhyolite flow and hence their directions represent a single spot reading of the geomagnetic field. The three sites are thus combined for a single virtual geomagnetic pole (VGP, Table 2). Bertrab directions are indistinguishable from Littlewood directions at the 5% significance level. This is consistent with Bertrab granophyre and Littlewood rhyolite being coeval, as demonstrated by their U-Pb ages. The combined Bertrab and Littlewood data sets (six sites) yield a group-mean VGP at 22.9°S, 260.3°E with an A_{95}=6.8° (Table 2). The angular dispersion of the VGPs is 8.1° with upper and lower 95% confidence limits of 14.2° and 5.7° [*Cox*, 1969]. The expected dispersion for the paleolatitude of the sampling area (28°) is about 14° for the present geomagnetic field and ranges from 9° to 18° over the past 200 Myr [*McFadden et al.*, 1991]. The error limits of these model calculations are about 2°. Results from Keweenawan rocks suggest that secular variation at circa 1.1 Ga was similar to current variation [*Halls and Pesonen*, 1982]. We regard the overlap, within the confidence limits, of the observed and expected dispersions as an indication that paleosecular variation has been averaged and that the combined data sets represent a time-averaged pole position. We nevertheless consider the alternative view that these data do not average paleosecular variation in discussions of paleogeographic reconstructions below.

Age of Magnetization

The high unblocking temperatures of the characteristic magnetizations of the rhyolite and granophyre would require a thermal event of similar temperature to remagnetize the samples [*Pullaiah et al.*, 1975]. Isotopic and petrographic studies clearly imply that such an event has not occurred since the crystallization and cooling of titanite at circa 1106 Ma. Chemical remagnetization by a low-temperature fluid event would require precipitation of two different magnetic minerals in the two rock units, with both minerals precipitating a very narrow range of grain size, as indicated by the narrow unblocking temperature spectra, two requirements that seem unlikely to result from a single event. These considerations imply that the magnetization is primary and, because of rapid cooling, has an age that is closely approximated by the U-Pb titanite age. Further evidence that the primary magnetization has been successfully isolated is the observation that our Coats Land pole is distinctively different from younger poles from East Antarctica (see Figure 10) [*Grunow*, 1995].

Structural disturbance

The very limited outcrop area precludes fully determining whether this region has been tectonically disturbed. There is, however, no field or microscopic evidence for deformation and the shallow dips of Littlewood rhyolite layering, as discussed above, are consistent with a depositional orientation. Moreover, the agreement between Littlewood and Bertrab poles suggests that disturbance at the outcrop scale has been minimal and implies structural coherence at a 10 km scale.

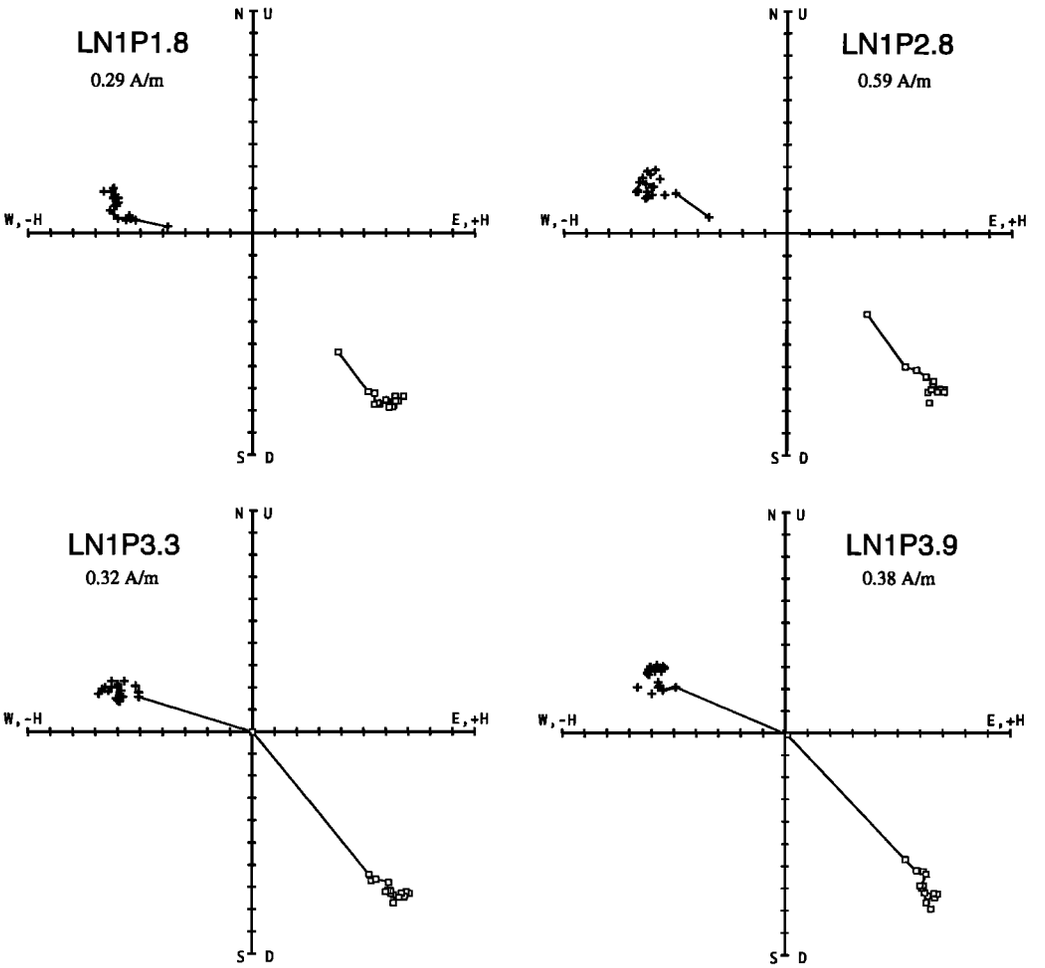


Figure 8. Vector component diagrams for samples of Littlewood rhyolite. Most samples retained >80% of their NRM intensity up to 660°C. Samples were thermally demagnetized at 150°, 200°, 250°, 300°, 350°, 400°, 450°, 500°, 525°, 550°, 560°, 570°, 580°, 600°, 620°, 640°, 660°, and 680°C. Conventions as in Fig. 6.

To evaluate structural disturbances at a larger scale requires comparison with other Precambrian poles from Antarctica, of which there are only two. *Hodgkinson* [1989] sampled 44 sites in intrusions of the Borgmassivet in the Grunehogna Province of western Dronning Maud Land (Figure 1), some 1200 km to the northeast of our sampling area. The sampling sites were distributed over approximately 2000 km^2. Because of the uniform response of pilot samples to stepwise alternating field demagnetization, bulk demagnetization levels were chosen for the remaining samples. No magnetic reversals were observed. A mean pole of 233°E, 6°S, A_{95}=7°was determined. Common directions from intrusions showing clearly different relative ages in the field led Hodgkinson to suggest that the rocks were remagnetized by a low temperature metamorphic event acting over a 10 to 100 m.y. time span. A 1200 Ma age of magnetization was arrived at by comparison with the African apparent polar wander path, in particular with the pole position from the Umkondo Lavas. *Hargraves et al.* [1994] reviewed the Umkondo and other pole positions from the Kalahari Craton and their age information, arriving at an average age of 1100 Ma for all of these African rocks. We note that *Moyes et al.* [1995] conclude, on the basis of Rb-Sr and Sm-Nd whole rock data, that the Borgmassivet intrusions

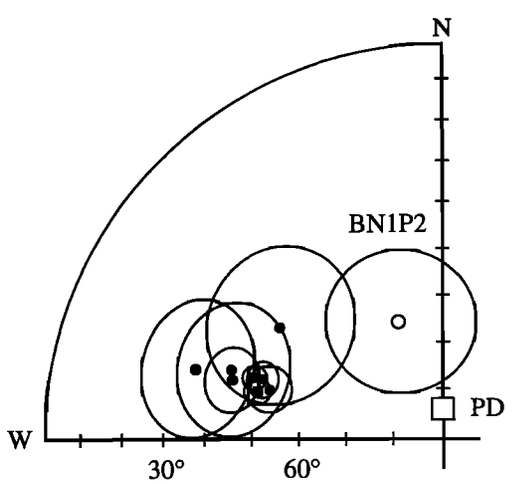


Figure 9. Site mean directions of Littlewood (squares) and Bertrab (circles) sites with their circles of 95% confidence. PD is the present dipole field direction in the sampling area. Bertrab directions are indistinguishable from Littlewood directions at the 5% significance level. The results from site BN1P2 have been deleted from the statistics.

Table 2. Statistical Parameters of Paleomagnetic Results

	N	*R*	*D*	*I*	*k*	α_{95}	Lat	Long	*dp*	*dm*
BN1P1	9(1)	8.94	285.7	52.6	128.3	4.6	29.1	78.1	4.3	6.3
BN1P3	9(1)	8.89	285.7	44.2	73.2	6.1	22.1	76.2	4.8	7.6
BN1P4	9(1)	8.53	286.2	35.5	17.0	12.9	15.8	75.2	8.6	14.9
BN2P1	9(1)	8.54	288.5	43.4	17.6	12.6	21.0	78.7	9.8	15.7
BN2P2	4(0)	3.92	304.8	48.7	35.7	15.6	22.3	94.9	13.5	20.5
LN1P1	10(0)	9.98	284.6	50.1	367.1	2.5	27.2	76.7	2.3	3.4
LN1P2	10(0)	9.98	288.0	48.9	472.3	2.2	25.5	79.6	1.9	2.9
LN1P3	10(0)	9.96	288.0	50.6	201.5	3.4	26.9	70.0	3.1	4.6
LN Sample Mean	30	29.91	286.9	49.9	304.6	1.5	26.5	78.8	1.3	2.0
LN Site Mean	3	3.00	286.9	49.9	2782.3	2.3	26.5	78.8	3.1	3.1
BN+LN POLES	6	5.95			98.8	6.8	22.9	80.3		
BN1P2	10	9.22	340.1	-64.1	11.5	14.9	57.1	299.6	19.0	23.8

N is the number of samples included (deleted) in the statistics; *R* is the resultant vector; *D* and *I* are the east declination and inclination; *k* is Fisher's precision parameter; α_{95} is the radius of the 95% circle of confidence; Lat and Long are the latitude and east longitude of the poleposition transferred ino the northern hemisphere; *dp* and *dm* are the semiaxes of the ellipses of 95% confidence. The BN (Bertrab) sites are located at 77° 52.4'S, 34° 38.1'W, the LN (Littlewood) sites at 77° 53.0'S, 34° 19.3'W.

have an age of circa 1000 Ma. It is unclear whether *Moyes et al.* [1995] dated any of the rocks used by *Hodgkinson* [1989].

In the Ahlmannryggen, also in the Grunehogna Province of western Dronning Maud Land (Figure 1), *Peters* [1989] collected samples from 11 flows and sills. He performed detailed geochemical, isotopic, petrographic, and rock magnetic analyses. Paleomagnetic samples were subjected to progressive demagnetization, mainly by alternating field, and characteristic directions were obtained by analyzing vector component diagrams. All sites were of the same polarity, which *Peters* [1989] interpreted to indicate that the sites are approximately the same age. A mean pole position of 239.9°E, 8.8°S, A_{95}=5.5° was determined. *Peters* [1989] considered a K-Ar age of 1183 ± 33 Ma from one dike to be representative for

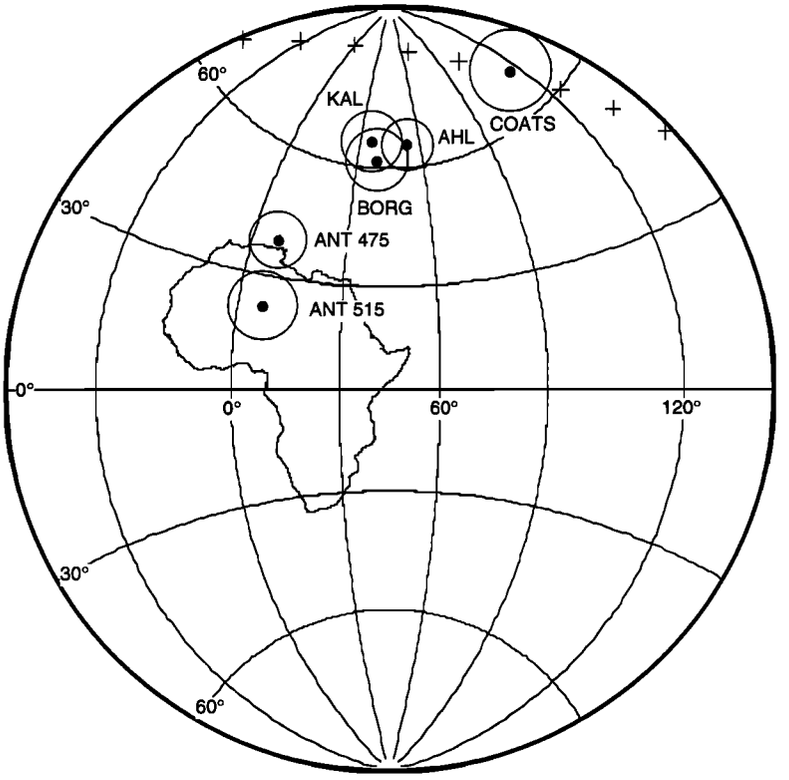


Figure 10. Comparison of mean paleomagnetic pole position of ~1100 Myr old rocks from the Kalahari craton of Africa (KAL) with poles from the Coats Land (COATS), Borgmassivet (BORG), and Ahlmannryggen (AHL). The poles from Antarctica have been rotated into the African coordinate system (see Table 3). Also shown are the mean Antarctica poles for 515 and 475 Ma [*Grunow*, 1995] in African coordinates. The trace of crosses shows, in 10° increments, the pole position of Coats Land if the sampling sites are subjected to vertical axis rotation. Stereographic projection.

all sampled rocks. The Borgmassivet and Ahlmannrhyggen poles are similar to ours (Figure 10), although large uncertainties in their age do not permit rigorous comparison.

Tectonic disturbances by undetected local vertical axis rotation are a potential source of the 30° angular difference between the Coats Land and Dronning Maud Land poles. The rocks nearest the Littlewood and Bertrab nunataks in the southern Maudheim Province are transected by a major Grenville dextral shear zone [*Jacobs*, 1991; *Jacobs et al.*, 1993]. If the shear zone extends southward beneath the ice, the Coats Land Nunataks would lie close to its projected trace. Strike-slip motion on this fault could have potentially resulted in local block rotations affecting the nunataks. Calculated pole positions for different amounts of hypothetical vertical axis rotations (Figure 10) do not, however, place the Coats Land pole appreciably closer to the other Antarctic poles, or to African poles (see below) for the same time period (Figure 10). Undetected tilt could also be a factor but restoration of the 10-11° dip at Littlewood Nunataks by rotation around the strike of the layering moves the Coats Land pole 2° away from the Dronning Maud Land poles and results in a slightly poorer grouping of the combined Littlewood and Bertrab data sets. To achieve overlap with the other Antarctic poles by tilting requires a minimum tilt of 31° around an axis nearly perpendicular to the inferred trace of the shear zone. There is no field evidence to support such a deformation.

We conclude from the above analysis that within the admittedly rather poor constraints of available data, large-scale tectonic disturbance is probably not the source of the disparity between the Coats Land and Dronning Maud Land poles. We thus infer that the differences are principally a consequence of differing ages of magnetization. Failure of the Coats Land pole to average paleosecular variation could also be a factor but, as noted above, the overlap of predicted and observed angular dispersions of the Coats Land VGPs implies this is not the case.

Discussion

In light of established geologic ties between Dronning Maud Land and southern Africa [*Groenewald et al.*, 1991, *Groenewald* 1993; *Jacobs et al.*, 1993], it is informative to compare the poles from Antarctica with poles of similar age from Africa after restoring the two cratons to their well-determined Gondwanan configuration [*Lawver et al.*, 1992]. There are five poles with an age of about 1100 Ma from the Kalahari craton that yield a mean pole at 65.4°N, 36.1°E and an A_{95}=6.5° [*Hargraves et al.*, 1994]. In this reconstruction (Figure 10), the Coats Land pole differs from the mean Kalahari pole by 30°. The two poles are statistically distinct, but this may be due to poor age constraints for the Kalahari poles. The Umkondo dolerites have a Rb/Sr whole rock-biotite isochron age of 1080 +140/-25 Ma [*Allsopp et al.*, 1989], the age of the post-Watersberg dolerites is derived from one sill dated by Rb/Sr at 1090 ± 15 Ma [*Allsopp et al.*, 1989], and the Timbavati gabbro has $^{40}Ar/^{39}Ar$ ages of 1074 ± 4 Ma and 1123 ± 5 Ma [*Burger and Walravem*, 1979, 1980]. Thus the difference between the Coats Land and the Kalahari poles may be due to their different ages and plate motion. In support of this possibility, we note that eight poles from North America with U-Pb ages between 1087 Ma to 1109 Ma (see Table 3) differ by up to 30°. We thus view the discrepancy between the two data sets as being largely due to poor age control and a limited data base and conclude that Coats Land, the Grunehogna Province and Kalahari craton could have been part of the same plate during this time interval. Failure of the Coats Land pole to average paleosecular variation could also account for the discrepancy of the two data sets but, as argued above, we believe this is not the case.

Figure 11 shows the apparent polar wander path (APWP) for North America in present coordinates for the time from 1200 to 1000 Ma based on the compilations of *Dunlop and Stirling* [1985] and *Costanzo-Alvarez et al.* [1993]. The path shown is

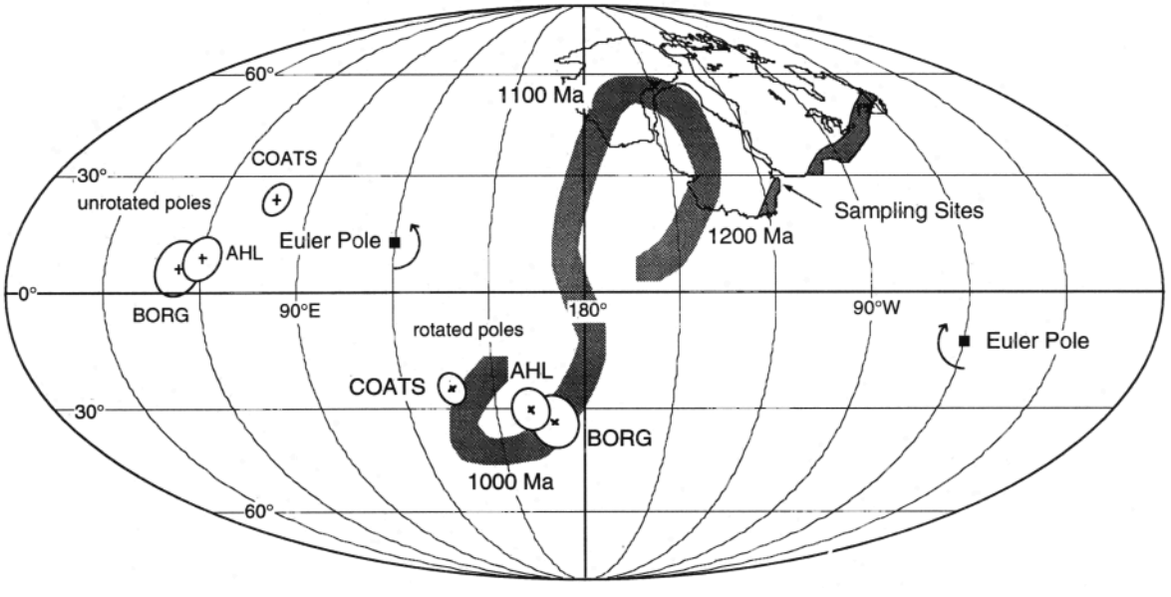


Figure 11. Generalized trend of the North American apparent polar wander path from ~1200 to ~1000 Ma and the Mesoproterozoic Antarctica poles (Coats Land (COATS), Ahlmannryggen (AHL), Borgmassivet (BORG)) in present coordinates and after rotation around the Euler pole for the SWEAT reconstruction. Antarctica and Australia are shown in the SWEAT configuration [*Dalziel*, 1991]. Dark shading indicates the Grenville Province of North America and its proposed continuation into East Antarctica [*Dalziel*, 1991]. Mollweide projection.

a schematic, free-hand fit meant only to show the general age trend; detailed, annotated paths are presented by *Harlan et al.* [1994] and *Park and Gower* [1996]. Also plotted is our mean pole position from southern Coats Land and the two Proterozoic poles from western Dronning Maud Land in present-day coordinates, as well as in their positions after juxtaposing East Antarctica with Laurentia as proposed by *Dalziel* [1991]. The rotated pole positions are derived from his SWEAT reconstruction, which is based on the match of the piercing points formed by the apparent truncation of the Grenville belts in Laurentia and East Antarctica. The reconstruction defines an Euler pole at 13°S, 300°E and a rotation angle of Ω =-135°. In present-day coordinates, the Antarctic poles are very different from the North American data, whereas in the SWEAT reconstruction, they fall on the 1000 Ma segment of the North American APWP. Although the overlap of the Coats Land pole with the North American APWP appears to be a positive result, its position is more than 60° away from the circa 1100 Ma part of the path, where the pole should fall on the basis of the isotopic ages reported above. Neither vertical axis rotation of the sampling area nor any reasonable tilt can bring the Coats Land pole significantly closer to the 1100 Ma poles of North America in the SWEAT fit. Likewise, failure to average secular variation can not alone explain the discrepancy; the angular distance between our pole and the North American poles of the same age is far in excess of what could be ascribed to such an error.

The age of magnetization for the Borgmassivet and Ahlmannryggen poles is not well enough determined to definitively argue how their pole positions relate in detail to the North American APWP, but at face value, they also do not fall on the APWP where they should according to the SWEAT reconstruction. These data thus seem to imply either that the SWEAT hypothesis is incorrect or that one major constraint for the reconstruction, the Grenville front correlation, is in error. Given the wide range of supporting data that have been offered since the hypothesis was first suggested [*Ross et al.* 1992; *Stump*, 1992; *Powell et al.*, 1993; *Borg and DePaolo*, 1994; *Ernst et al.*, 1995; *Park et al.*, 1995; *Young*, 1995] and the wealth of global observations that juxtaposition of Laurentia and East Antarctica-Australia seems to explain (see summary by *Dalziel* [1995]), we deem the latter possibility more likely and explore it below.

A principal tenet of the piercing point correlation of *Dalziel* [1991] was that the circa 1.1 Ga rocks of the Coats Land Nunataks and those of the Maudheim and Grunehogna provinces to the north in western Dronning Maud Land were part of the East Antarctic craton by the end of Grenvillian oro-

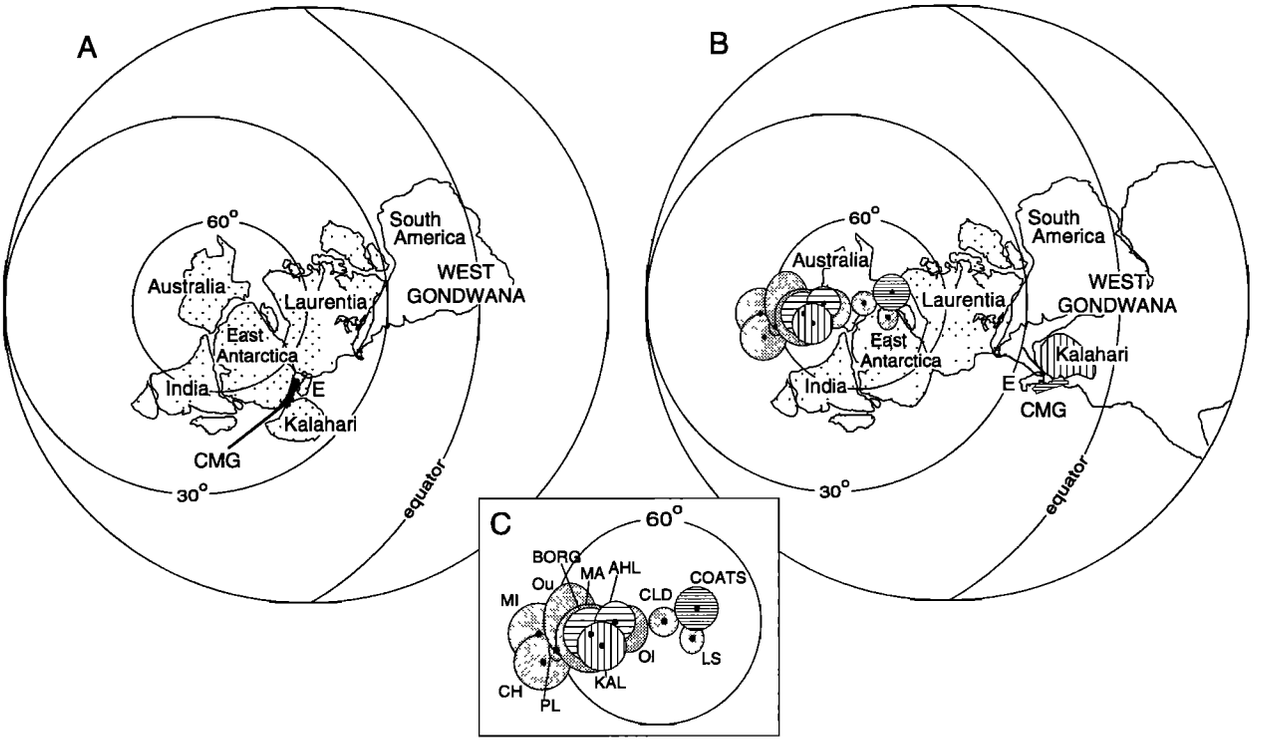


Figure 12. (a) Rodinia reconstruction as proposed by *Dalziel* [1991, 1992a]. The Coats Land-Maudheim-Grunehogna (CMG) province, the Ellsworth-Whitmore Mountain Block (E), and the Kalahari craton of Africa are part of East Gondwana (stippled). Stereographic projection. (b) Proposed plate reconstruction for ~1100 Ma that makes the Coats Land-Maudheim-Grunehogna (CMG) province, the Ellsworth-Whitmore Mountain Block (E), and the Kalahari craton part of West Gondwana. Geological justification for this reconstruction is given by *Dalziel* [1992b, 1994]. In this scenario, the Coats Land, Ahlmannryggen, Borgmassivet poles from Antarctica and the mean Kalahari pole fall precisely onto the Laurentian apparent polar wander path of the corresponding age. The continents are shown with their present coast lines for ease of recognition only. Stereographic projection. (c) Identification of paleomagnetic poles which are shown with their 95% confidence errors. COATS, Coats Land; AHL, Ahlmannryggen; BORG, Borgmassivet; KAL, Kalahari; LS, Logan Sills; CLD, Coldwell; Ol, lower Osler; Ou, upper Osler; PL, Portage Lake; MI, Michipicoten; MA, Mamainse Point; CH, Copper Harbor. See Table 3 for references and ages.

genesis. On the basis of studies that linked the geology of western Dronning Maud Land to that of southern Africa [*Groenewald et al.*, 1991; *Moyes et al.*, 1993; *Jacobs et al.*, 1993], this in turn implied that the Kalahari craton of Africa was part of East Gondwana, as the Archean and Proterozoic rocks of the Grunehogna Province on the continental margin of East Antarctica were clearly part of that craton prior to the Mesozoic opening of the southwest Indian Ocean. Rodinia reconstructions of *Dalziel* [1992a, b, 1995] (Figure 12a) reflect this premise. Our new data suggest that it is incorrect.

The structural relationship of the Coats Land Nunataks to the Grunehogna Province cannot be directly established, as the two are separated by over 1000 km of the East Antarctic ice sheet (Figure 1). The general agreement of the Coats Land pole with the poles from the Borgmassivet and Ahlmannryggen is permissive of the idea that within present-day Antarctica, Coats Land and western Dronning Maud Land were part of a single tectonic unit along the margin of the present Weddell Sea by about 1.1 Ga. We refer to this terrain herein as the Coats Land-Maudheim-Grunehogna (CMG) province. The lack of overlap of the CMG poles with poles of the same age on the North American APWP after restoration to the SWEAT reconstruction (Figure 11) suggests, however, that the CMG province may not have been part of the East Antarctic craton during the same time period. This implies that the contact between the CMG and older rocks of the East Antarctic craton is younger than 1.1 Ga and not Grenvillian in age, an inference supported by recent suggestions of a continuation of the Neoproterozoic to Cambrian(?) East African orogenic belt into East Antarctica and the Shackleton range [*Shiraishi et al.*, 1994; *Grunow et al.*, 1996; *Tessensohn*, 1996]. This orogenic belt is believed to mark the suture between East and West Gondwana that formed during the final amalgamation of Gondwanaland [*Dalziel*, 1991; *Hoffman*, 1991; *Stern*, 1994]. Given the established geologic ties between the Kalahari craton and the CMG, the proposed continuation of the suture into the Shackleton range offers the possibility that the CMG was part of West Gondwana, not East Gondwana, prior to suturing with the East Antarctic craton in latest Precambrian to Cambrian time.

Figure 12b shows a late Proterozoic plate reconstruction that could explain our paleomagnetic data from Coats Land. This reconstruction is broadly based on the suggestion of *Dalziel* [1991] and *Hoffman* [1991] that Laurentia was located in Rodinia between East and West Gondwana during Neoproterozoic times following Grenville orogenesis. Specifically, it shows the Kalahari craton in the position with respect to Laurentia that was suggested by *Dalziel* [1992b, 1994] for the hypothetical latest Neoproterozoic supercontinent, Pannotia, following opening of the Pacific Ocean and amalgamation of Gondwanaland along the East African suture. The Coats Land-Maudheim-Grunehogna (CMG) province and Ellsworth-Whitmore Mountains block of Antarctica are attached to the Kalahari craton as part of West Gondwana, rather than being attached to the East Antarctic craton and hence part of East Gondwana. We highlight the Kalahari craton and the CMG terrane in the reconstruction, leaving the remainder of West Gondwana undifferentiated. This is because the timing of suturing along the latest Precambrian to Cambrian Pan-African and Brazilide orogenic belts of West Gondwana remains problematic and is certainly in part younger than 1100 Ma [see, e.g., *Renne et al.*, 1990; *Hoffman*, 1991; *Dalziel*, 1992b; *Grunow et al.*, 1996].

Also shown in Figure 12b are eight poles from North America ranging in age from 1087 to 1109 Ma as well as the three Antarctic poles and the mean Kalahari pole, the Antarctic and Kalahari poles having been rotated corresponding to the paleogeographic reconstruction (Table 3). In this scenario, the Coats Land pole falls precisely on the North American apparent polar wander path and the Borgmassivet, Ahlmannryggen, and Kalahari poles lie on the same path displaced toward slightly younger ages. This observation is sup-

Table 3. Mesoproterozoic Paleomagnetic Data From Antarctica, Africa, and Laurentia

Rock Unit	Reference Frame							
	Antarctica		Africa		Laurentia			
	Lat	Long	Lat	Long	Lat	Long	Reference	Age, Ma[a]
Coats Land	22.9°N	80.3°E	68.0°N	116.3°E	58.7°N	216.6°E	this study	
Ahlmanryggen	8.8°N	59.9°E	64.7°N	54.1°E	42.1°N	188.4°E	*Peters* [1989]	
Borgmassivet	6.0°N	53.0°E	61.4°N	39.4°E	35.2°N	185.2°E	*Hodgkinson* [1989]	
Kalahari Mean			65.4°N	36.1°E	34.7°N	190.4°E	*Hargraves et al.* [1994]	
Logan Sills					49.4°N	219.9°E	*Halls and Pesonen* [1982]	1109
Coldwell					50.4°N	204.5°E	*Harlan* [1993] [b]	1108
Upper Osler					33.9°N	177.9°E	*Halls* [1974]	<1098
Lower Osler					43.1°N	194.5°E	*Halls* [1974]	1098
Portage Lake					26.5°N	181.2°E	*Halls and Pesonen* [1982]	1095
Michipicoten					26°N	175°E	*Palmer and Davis* [1987]	<1087
Mamainse Point					38°N	186°E	*Palmer and Davis* [1987]	>1087
Copper Harbor					22.2°N	180.8°E	*Diehl and Haig* [1994]	1087

Euler poles are Antarctica to Africa: 7.1°N, 144.8°E, Ω=-56.4° [*Lawver et al.*, 1992]; Africa to South America: 50.0°N, 327.5°E, Ω=-55.1° [*Nürnberg and Müller*, 1991]; South America to Laurentia: 2.3°N, 336.4°E, Ω=-99.3° [*Dalziel et al.*, 1994].

[a] U-Pb age determinations. Portage Lake age by *Davis and Paces* [1990].

[b] After *Lewchuck and Symons* [1990a].

portive of our earlier suggestion that the angular separation between the Coats Land pole and the Kalahari poles is due to age differences.

Conclusions

U-Pb zircon and titanite ages for the volcanic and hypabyssal rocks of the Coats Land Nunataks, East Antarctica indicate emplacement, crystallization and cooling of this coeval suite between 1112 ± 4 Ma and 1106 ± 3 Ma. Well-preserved primary igneous textures and structures in all exposures, combined with the absence of deformational fabric or significant lead loss indicates that rocks of the Coats Land Nunataks have not undergone any thermal disturbance or penetrative deformation since that time. Paleomagnetic analyses from all sampled rock types in the nunataks yield statistically indistinguishable results that give a group-mean pole position of 22.9°S, 260.3°E with an A_{95}=6.8°. This pole compares favorably with the only other extant Precambrian paleomagnetic data for East Antarctica, two less well-dated poles from the Grunehogna Province, some 1200 km northeast of the Coats Land exposures, and with comparable age poles for the Kalahari craton of southern Africa. Allowing for small differences in age and some plate motion, the agreement of the Coats Land VGP with those from western Dronning Maud Land and the Kalahari craton, as well as previously recognized links between the geology of western Dronning Maud Land and the Kalahari craton, favors the idea that the CMG province was attached to the Kalahari craton. The boundary between the CMG province and the East Antarctic craton, previously considered the along strike continuation of the Grenville Front into East Antarctica and a piercing point in the SWEAT reconstruction, is thus probably not a Grenville-age feature. Instead, it may mark the position of latest Precambrian to Cambrian suturing of the CMG province-Kalahari craton to the East Antarctic craton. This boundary can be viewed as a continuation of the East African suture between East and West Gondwana, which would then extend for approximately 7500 km, from Arabia to the Pacific margin of the East Antarctic craton. A new reconstruction that places the Kalahari craton and the CMG province within West Gondwana and off the present southeastern margin of Laurentia is consistent with poles of circa 1.1 Ga from Laurentia, Kalahari craton, and the CMG province.

Appendix: Analytical Methods

U-Pb Geochronology

U-Pb analytical work was performed at the University of Texas at Austin. Rock samples were crushed and minerals were separated under clean conditions using a rock crusher, disc pulverizer, Rogers table, sieves, heavy liquids and Frantz magnetic separator. Mineral fractions were selected for analysis on the basis of microscopic optical properties to ensure that only the highest quality grains were analyzed. All zircon and titanite fractions analyzed were strongly abraded, subsequently optically reevaluated and then washed successively in distilled 4 *N* nitric acid, water and acetone. They were loaded into Teflon capsules with a mixed $^{205}Pb/^{235}U$ isotopic tracer and dissolved with HF and HNO_3 acids. Chemical separation of U and Pb from zircon using minicolumns (0.055 mL resin volume [after *Krogh,* 1973] resulted in a total Pb procedural blank between 1-2 pg. Chemical separation from titanite on 0.5 mL columns resulted in a total procedural Pb blank of approximately 5 pg. U was 2 pg for both column types. Pb and U were loaded together with silica gel and phosphoric acid onto an outgassed rhenium ribbon and analyzed on a multicollector MAT 261 thermal ionization mass spectrometer, operating in static mode with ^{204}Pb measured in the axial secondary electron multiplier (SEM) - ion counting system. Ages were calculated using the decay constants of *Jaffey et al.* [1971]. Errors on the isotopic ratios in Table 1 are reported at 2σ. They were calculated by propagating uncertainties in measurement of isotopic ratios, fractionation and amount of blank with a program modified after an unpublished error propagation program written by L. Heaman (Royal Ontario Museum, Toronto, Canada). Linear regressions were calculated using the procedure of *Davis* [1982]. The goodness of fit of a regressed line is represented as a probability of fit, where 10% corresponds to a mean square of weighted deviates (MSWD) of 2. Ages given in the text, table, and figures are quoted with a 2σ error.

Paleomagnetism

Oriented samples were collected using a portable drill. Sample locations were determined by a hand-held global positioning system (GPS) receiver. At each site, the samples were collected over a 5 to 20 m^2 area in an attempt to average the secular variation of the geomagnetic field. In the laboratory, the samples were kept in a magnetically shielded room throughout the measuring procedure. The magnetic remanence was measured with a cryogenic magnetometer and all samples were subjected to progressive thermal demagnetization, 15 steps for Bertrab samples and 19 steps for Littlewood samples. The data were analyzed using principal component analysis [*Kirschvink,* 1980]; site- and group-mean directions were calculated using Fisher [1953] statistics.

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